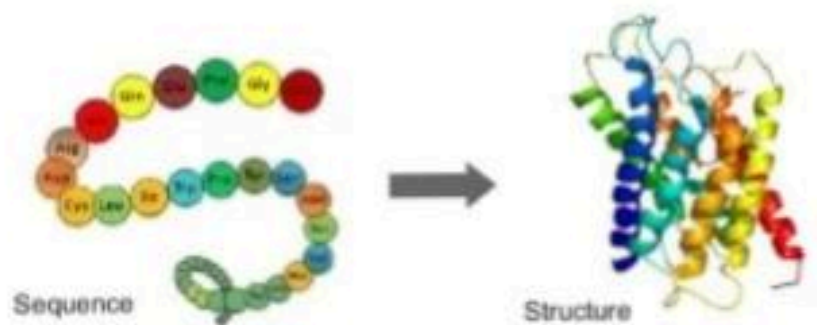


Protein

- ✓ Proteins are formed of amino acid residue (more than 100 amino acids) linked together by peptide bonds.
- ✓ They are of high molecular weight ,colloidal in nature , and heat labile.



Each protein has a **unique** amino acids sequence
Alteration in amino acid sequences result in abnormal function or diseases



Biomedical importance of proteins

I- Structural functions

- Essential building blocks of the cells
- Essential for the repair of damaged tissues.

II - Dynamic functions

- ☐ Catalytic function e.g. enzymes
- ☐ Regulation of metabolism e.g. hormones
- ☐ Movement function e.g. actin, tubulin
- ☐ Storage function
 - Ferritin stores iron .
 - Ceruloplasmin stores copper

☐ Defense or protective function

- **keratin** found in skin cells protects against mechanical and chemical injury
- **Blood clotting proteins** fibrinogen and thrombin prevent blood loss
- Antibodies (Immunoglobulins)

☐ Transport function

- Na-K ATPase and glucose transporter
- Hemoglobin
- Lipoproteins
- Albumin carries calcium, free fatty acids and bile pigments.
- Transferrin carries iron

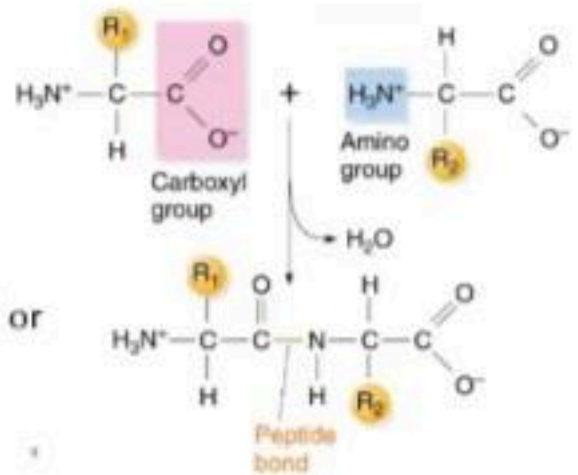
Protein structure

☐ Bonds responsible for protein structure are:

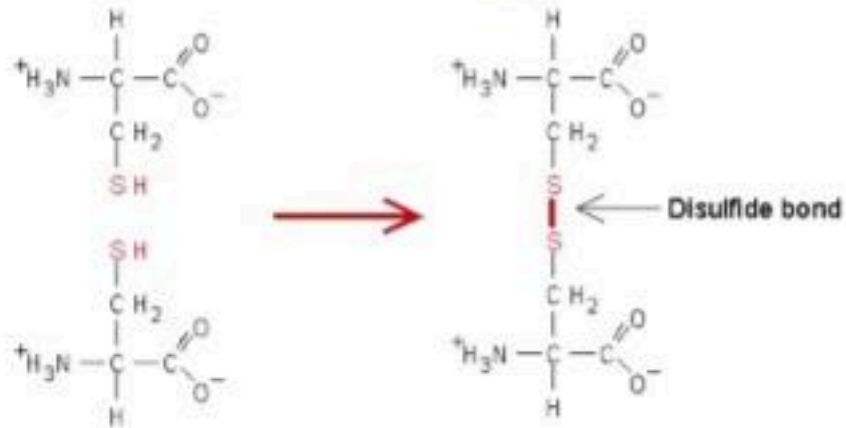
I- Covalent (strong) bonds

1. Peptide bond

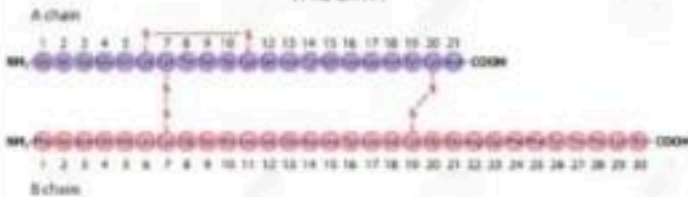
- It stabilizes the protein structure
- Prevent rotation of protein molecule
- broken only by enzymatic action or by strong acid or base at high temperature



2. Disulfide bond

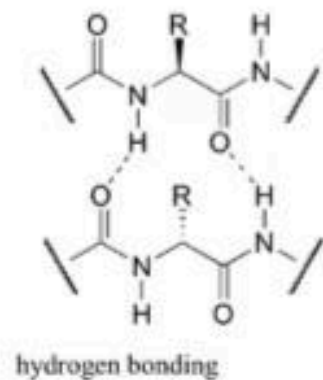
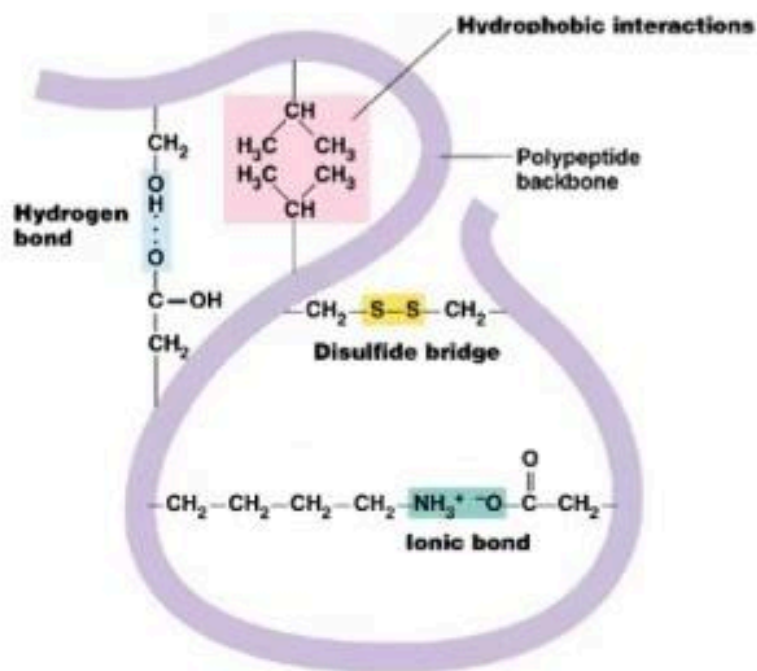


Insulin



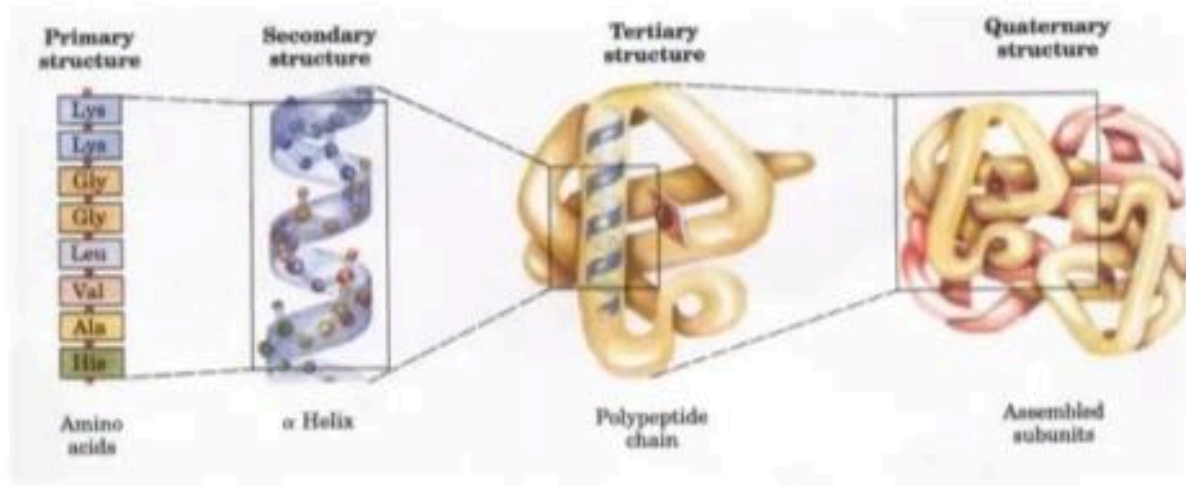
II- Non covalent or weak bonds

1. Hydrogen bonds
2. Hydrophobic interactions
3. Electrostatic bonds (ionic interaction or salt bridge)



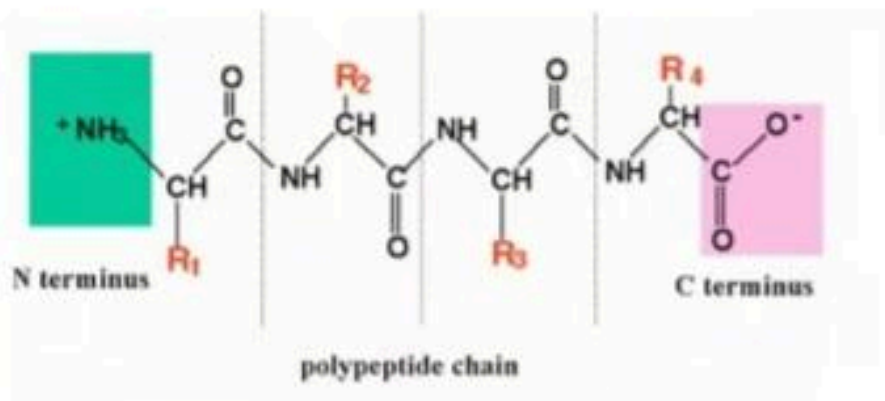
Conformation of proteins

- Protein molecule has a characteristic three dimensional shape
(**primary, secondary and tertiary** structure)
- Proteins formed of two or more polypeptide chains have **quaternary** structure



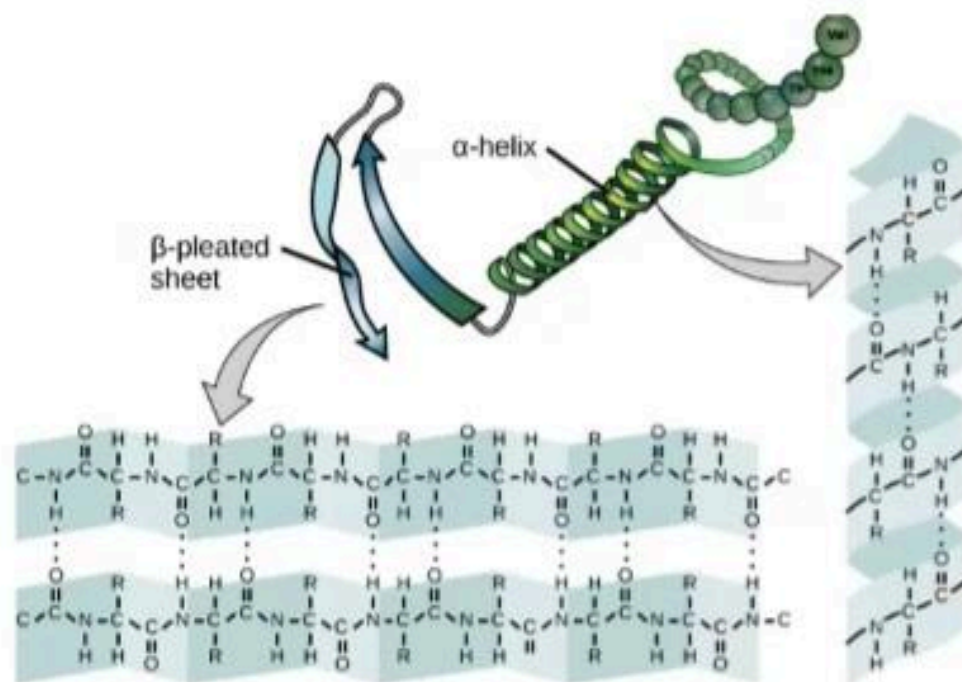
1. Primary structure of proteins

- Refers to the **number** and **sequence (order)** of amino acids in the polypeptide chain
- The free α -amino group, written to the left, is called the amino-terminal or **N-terminal end**.
 - The free α -carboxyl group, written to the right, is called the carboxyl-terminal or **C-terminal end**.



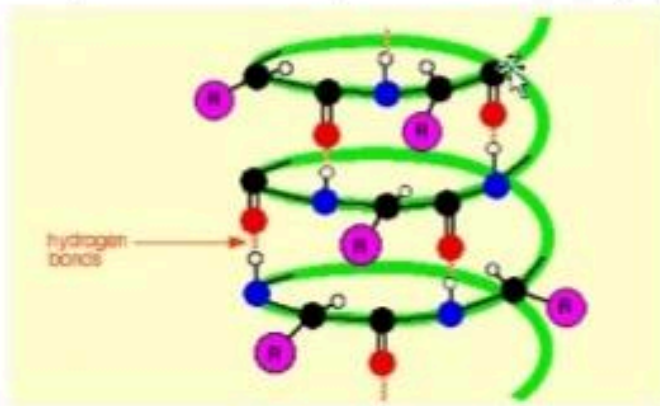
2. Secondary structure of proteins

- Refer to Coiling, folding or bending of the polypeptide chain
- The most common types of secondary structures are **α helix** and the **β pleated sheet**
- These secondary structures are held together by **hydrogen bonds**.

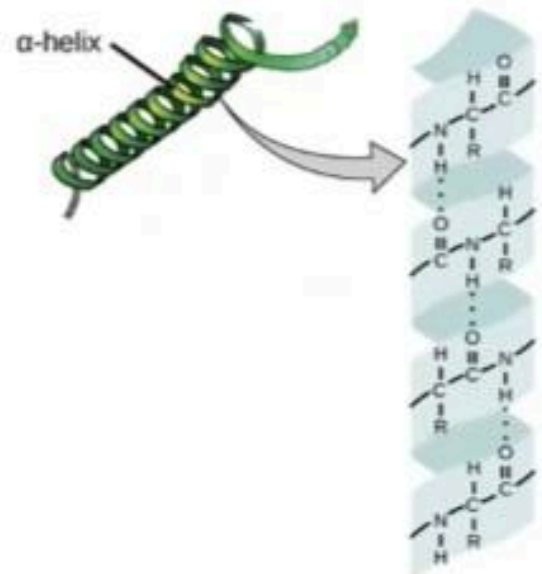


α -helix

- Is a right-handed helix that is held together by hydrogen bond
- The hydrogen bond is formed between the N-H on one amino acid and the C=O on another amino acid 4 residues away
- each turn of the helix containing 3.6 amino acids.
- typical α -helix is about 11 a.a.
- Examples of α -helical protein: keratin, Myoglobin

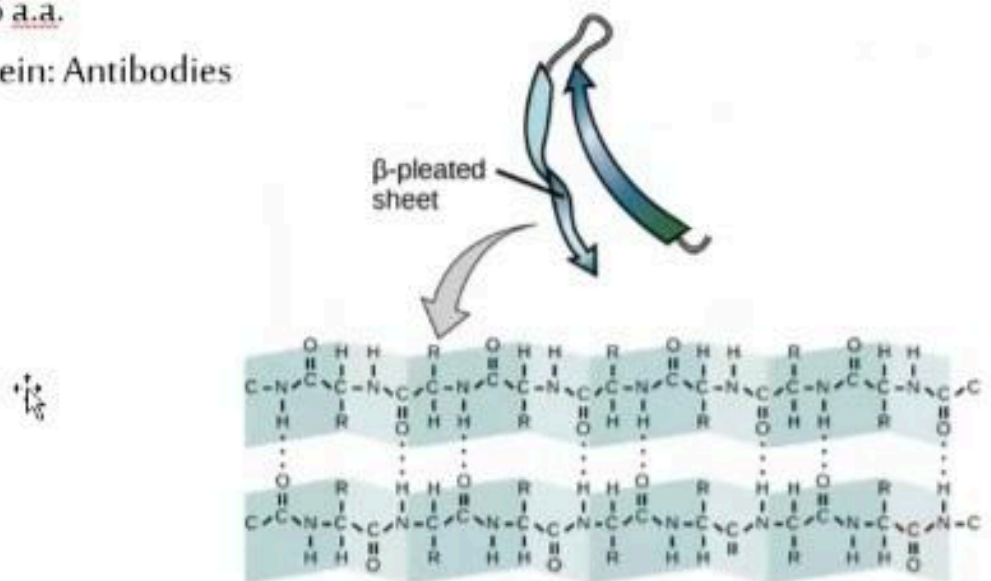


N.B. α -helix in **collagen** are left-handed



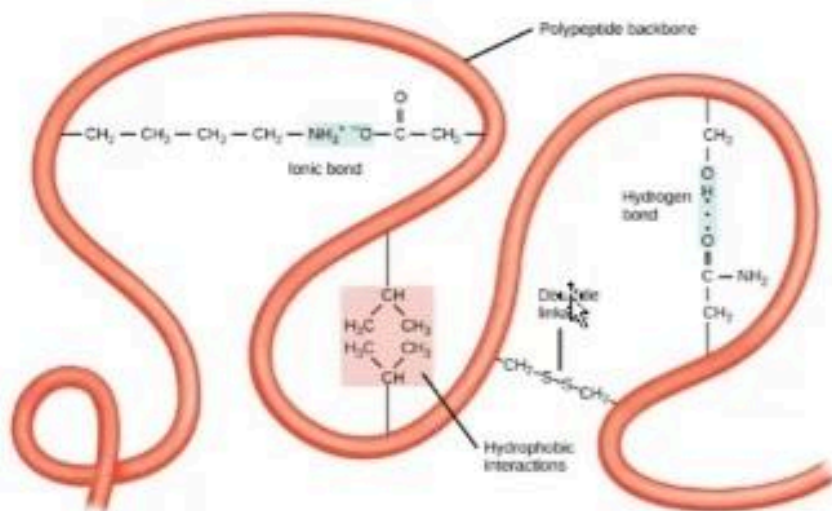
β pleated sheet

- two or more segments of a polypeptide chain line up next to each other, forming a sheet-like structure held together by hydrogen bonds
- In contrast to the alpha helix, hydrogen bonds in beta sheets form in between N-H groups in one strand and C=O groups in the adjacent strands
- Typical β sheet is about 6 a.a.
- Examples of β -sheet protein: Antibodies



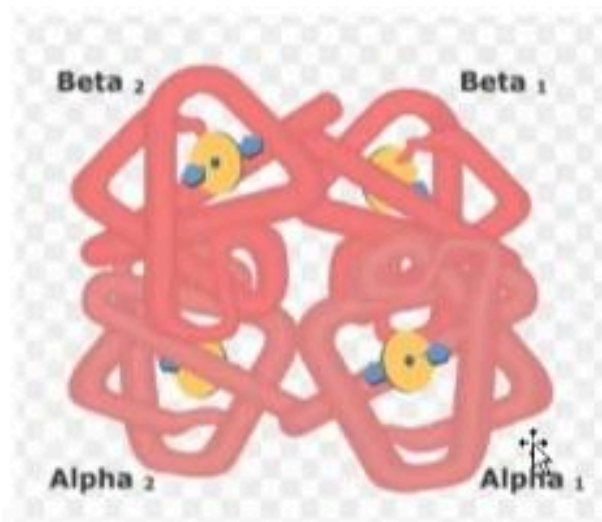
3. Tertiary structure of proteins

- It is the three dimensional structure of each polypeptide chain
- Bond stabilizing tertiary structure **include** disulfide bonds, hydrophobic interactions, hydrogen bonds and ionic interactions
- two main forms of tertiary structure; **fibrous** and **globular** types



4. Quaternary structure of proteins

- It is the arrangement of a protein that has more than one polypeptide chain (oligomers).
- Each polypeptide chain in such a protein is called **monomer or subunit**
- Hemoglobin is an example of protein present in quaternary structure. It is tetramer having two α chains and two β chains.



Properties of proteins

1. Solubility:

- Soluble in water e.g. albumin
- Soluble in salt solution e.g. globulin
- Soluble in alcohol e.g. protamine
- insoluble at all e.g. scleroproteins

2. Amphoteric property.

- they can behave as acids and base (similar to a.a.)
- Therefore acting as buffers

3. Isoelectric point of proteins

It is the pH at which the protein carries equal positive and negative charges (zero charge)

At IEP

- ✓ The proteins do not move in electric field
- ✓ Proteins lose their physiological functions
- ✓ Viscosity of protein solution is minimal
- ✓ Easily precipitated and denatured

4. Color reactions:

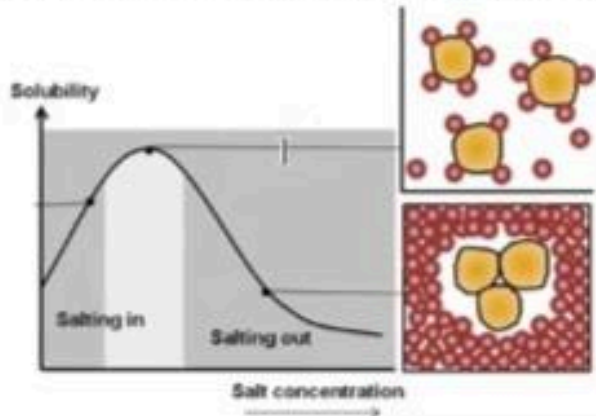
All color reactions given by amino acids can be given by protein containing these a.a.

5. Salting in and salting out

Salting in: adding small amounts of salts (such as sodium chloride) to protein solution increases the solubility of the protein due to increasing the ionic strength of protein

Salting out: adding large amounts of salts (ammonium sulfate) to protein solution result in their precipitation because salt ions compete with the proteins for water molecules.

- ✓ The salt concentration at which a protein precipitates differs from one protein to another
- ✓ It is reversible process



ing the electrostatic energy between the protein molecules which increase the solubility of proteins.

Denaturation of proteins

- It is loss of native structure (natural conformation) of protein
- Denaturation disrupts all orders of protein structure (secondary, tertiary and quaternary) **except** primary structure
- non covalent bonds and may be disulfide but not peptide bonds are ruptured
- In many cases, denaturation is irreversible

Causes of protein denaturation

1. Physical agents: such as heating, vigorous shaking and stirring, repeated freezing and thawing, UVR and exposure to high pressure.
2. Chemical agents: urea, salts of heavy metals as mercury and lead disrupt ionic bonds, strong acids or bases, sulfhydryl agents as β -mercaptoethanol, alkaloidal reagents as picric acid and phosphotungstic acid and alcohol.

N.B. Denaturation can also be caused by changes in the pH

re or pH of a protein's environment is changed, or if it is exposed to chemicals, these interactions may be disrupted.

Effects of denaturation:

1- Physical changes

- Decreased solubility of protein
- Increased viscosity of proteins

2- Chemical changes

- Rupture of non covalent bonds .
- Exposure of some groups

3- Biological changes

- Loss of function
- Loss of biological activity of enzymes and protein hormones.
- Changes of antigenic property of proteins.

N.B.

- ✓ Denatured proteins (cooked foods) are easily digested
- ✓ Coagulation cause irreversible precipitation of globular proteins (albumin and globulins)

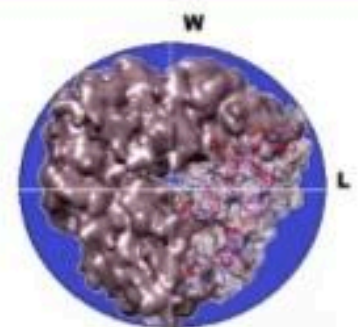
Classification of proteins

I- According to shape

proteins classified according to axial ratio into :

$$\text{Axial ratio} = \frac{\text{length}}{\text{width}} = \frac{L}{W}$$

	Fibrous proteins	Globular proteins
Axial ratio	more than 10	less than 10
Shape	long rod shaped	compact spherical
Examples	keratin, myosin, fibrin and collagen	albumin, globulin, and enzymes.



Classification of proteins

II- According to biological function

<i>Class of protein</i>	<i>Function</i>	<i>Examples</i>
Enzymic proteins	Biological catalysts	Urease, Amylase, Catalase, Cytochrome C, Alcohol dehydrogenase.
Structural proteins	Strengthening or protecting biological structures	Collagen, Elastin, Keratin, Fibroin
Transport or carrier proteins	Transport of ions or molecules in the body 	Myoglobin, Hemoglobin, Ceruloplasmin, Lipoproteins
Nutrient and storage proteins	Provide nutrition to growing embryos and store ions	Ovalbumin, Casein, Ferritin
Contractile or motile proteins	Function in the contractile system	Actin, Myosin, Tubulin
Defense proteins	Defend against other organisms	Antibodies, Fibrinogen, Thrombin
Regulatory proteins	Regulate cellular or metabolic activities	Insulin, G proteins, Growth hormone
Toxic proteins	Hydrolyze (or degrade) enzymes	Snake venom, Ricin.

Classification of proteins

III- According to the biological value:

1. **Proteins of high biological value**

contain all the essential amino acids e.g. animal proteins.

2. **Proteins of low biological value**

deficient in one or more essential amino acids e.g. plant proteins as zein.

IV- According to the chemical composition:

1. **Simple proteins** : consist of amino acids only.

2. **Conjugated proteins**: consist of amino acids and non protein or prosthetic groups.

3. **Derived proteins**: are hydrolytic products of simple and conjugated proteins.

Simple proteins

1- Albumin and globulins:

Albumin	Globulins
Soluble in water and salt solution. Coagulated by heat. Globular protein of high biological value contains most essential a. a. Easily digested. M.W: 68000 Daltons Precipitated by full saturation. It present in: Blood: serum albumin. Milk: <u>lactalbumin</u> . Egg: ovalbumin.	Soluble in salt solution. Coagulated by heat. Globular protein of high biological value contains most essential a. a. Easily digested. M.W: 150000 Daltons. Precipitated by half saturation. It presents in: Blood: serum globulins. Milk: <u>lactoglobulins</u> . Egg: <u>ovaglobulins</u> .

2- Protamines

- basic proteins rich in lysine and arginine
- bound to DNA in the nuclei of sperm forming nucleoproteins.
- They are soluble in water.

3- Histones

- basic proteins rich in histidine
- bound to DNA in the nuclei of plants and animals forming nucleoproteins
- soluble in water.

4- Gliadins (prolamines)

- acidic proteins rich in glutamic acid
- present in wheat, maize (corn) and rice
- Gliadin of the maize is called **zein** which lacks 2 essential amino acids(tryptophan and lysine)

5- Glutelins

- acidic proteins rich in glutamic acid, present in wheat, corn and rice
- They are heat coagulable proteins

6- Scleroproteins (connective tissue proteins or albuminoids)

- They are fibrous structural proteins
- insoluble in water, acids, alkalis or alcohol
- They have supportive and protective functions
- Include

1- Keratin (epidermal protein):

It is rich in sulfur containing amino acid cysteine and presents in two types:

- ✓ α -keratin: it is the protein of outer surface of the skin, hair and nails.
- ✓ β -keratin: it presents in silk

2- Collagen

- It presents mainly in skin, cartilage, tendons and ligaments.
- They can be digested by pancreatic collagenase.
- On boiling, they are converted to gelatin.

3- **Elastin**: It is protein of elastic tissues (lung, wall of blood vessels and ligaments).

4- **Ossein**: It is protein found in bone and teeth.

5- **Reticulin**: It is the protein of reticular tissue being similar to collagen.

Conjugated or compound proteins

According to the prosthetic groups, they are classified into:

1- Phosphoproteins:

- They contain phosphoric acid as prosthetic group
- They are of animal origin e.g. caseinogens present in milk and vitellin present in egg yolk.

N.B. Caseinogens is converted by rennin to soluble casein which is precipitated by calcium as calcium caseinate (**cheese**).

2- Glycoproteins and Mucoproteins

- They contain carbohydrates as prosthetic group

3- Lipoproteins:

- These are combinations of proteins with lipids.
- They are present in cell membranes, plasma lipoproteins and thromboplastin

4- Nucleoproteins:

- They contain nucleic acids (DNA or RNA) as prosthetic group attached to protamines or Histones.
- They are found in cell nuclei and also in cytoplasm.

5- Chromoproteins

These are proteins that contain colored prosthetic groups such as:

- ✓ Hemoglobin, cytochromes, Catalase and peroxidases (contain hem pigment (red)
- ✓ Flavoproteins (contain riboflavin (yellow)).
- ✓ Rhodopsin (visual purple) is composed of opsin and 11-cis retinal.

6- Metalloproteins:

- They contain metals as prosthetic groups and include:
 - ✓ Hemoglobin and ferritin (contain iron).
 - ✓ Carboxypeptidase, insulin and carbonic anhydrase (contain zinc).

Derived proteins

- They are hydrolytic products of proteins
- Resulting from the action of heat, enzymes or chemical reagents
- According to molecular weight, they are classified into:
 1. Metaproteins.
 2. Proteoses.
 3. Peptones.

Gelatin :

- ✓ is a hydrolytic product of collagen.
- ✓ It is poor in essential amino acids
- ✓ used as a supplementary protein as it is easily digested.

